



Technical Service Bulletin: TSB210001

Released Date: 19-Feb-2024

New Thermocouple Exhaust Gas Temperature Sensor (EGTS) Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK50 CM2150 MCRS
- QSK50 CM2150 K107

Issue

Symptom:

- Exhaust gas temperature sensor (EGTS) Fault Codes: FC671, FC672, FC673, FC674, FC675, FC676, FC677, FC678, FC722, FC723, FC724, FC725, FC726, FC727, FC728, FC1521, FC1522, FC1523, FC1524, FC1525, FC1526, FC1527, FC1529, FC1532, FC1533, FC1534, FC1535, FC1536, FC1618, FC1619, FC2735, FC2736.
- No reading or incorrect EGTS signal to engine control module (ECM).

Root Cause:

- The exhaust gas temperature sensor (EGTS) cable tensile strength capability has been exceeded due to the sensor cabling being exposed to high heat causing cable fracture.
- Proximity of the EGTS to exhaust manifolds which have radiant temperature of greater than 400°C [752°F] causes the EGTS transition point (from the solid to the braid) to overheat and malfunction.

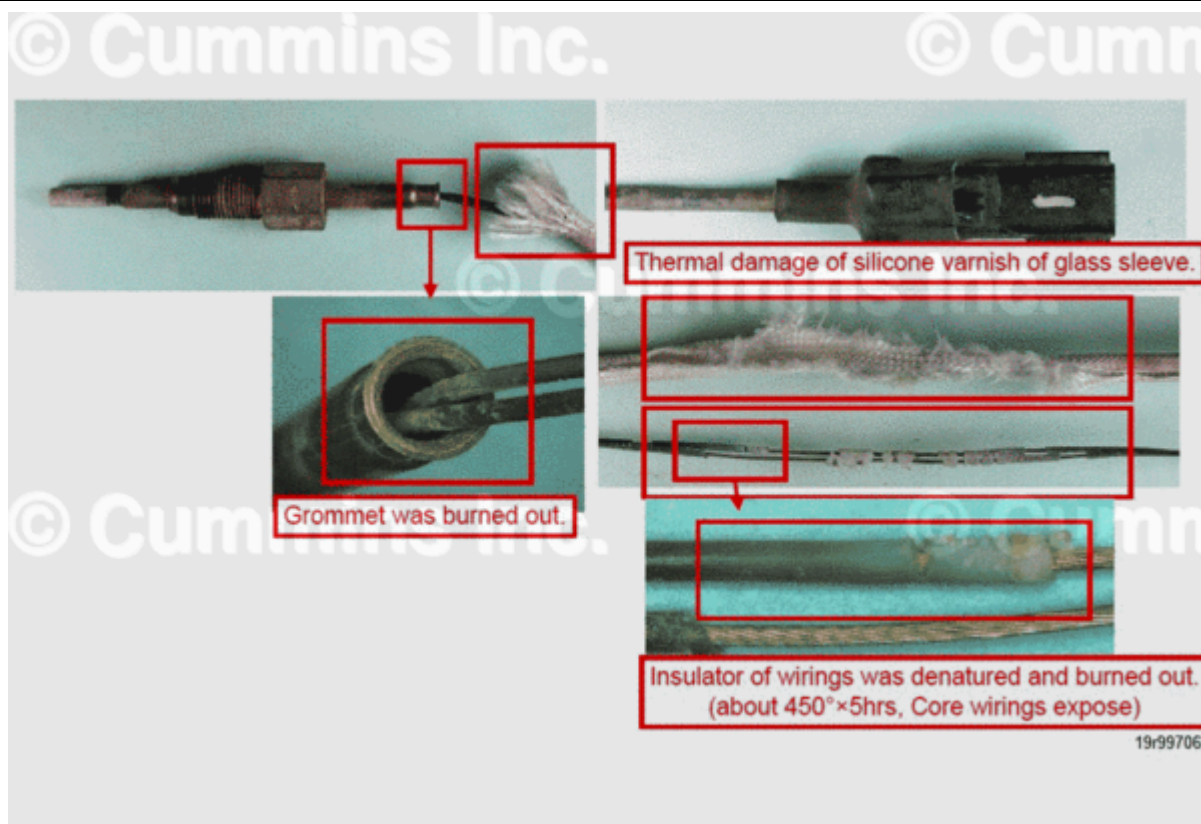


Figure 1, EGTS Damage Examples.

Verification

- Visual inspection of the EGTS cable will show thermal damage to the EGTS's cable i.e. exposed bare wires, chaffing.
- Use of a multimeter to measure across the terminals of the sensors results in extremely high resistance or open circuit.
- Use of Cummins® electronic service tool or equivalent to confirm presence of any of the fault codes listed above.

Resolution

- New longer body thermocouple EGTS, which have higher working temperature capability around the exhaust manifold area, have been released. Service Parts Availability section below for part numbers.
- Existing thermistor EGTS are **not** compatible with thermocouple EGTS. See Part Compatibility section below.

Description of Change

Existing Thermistor Exhaust Gas Temperature Sensors

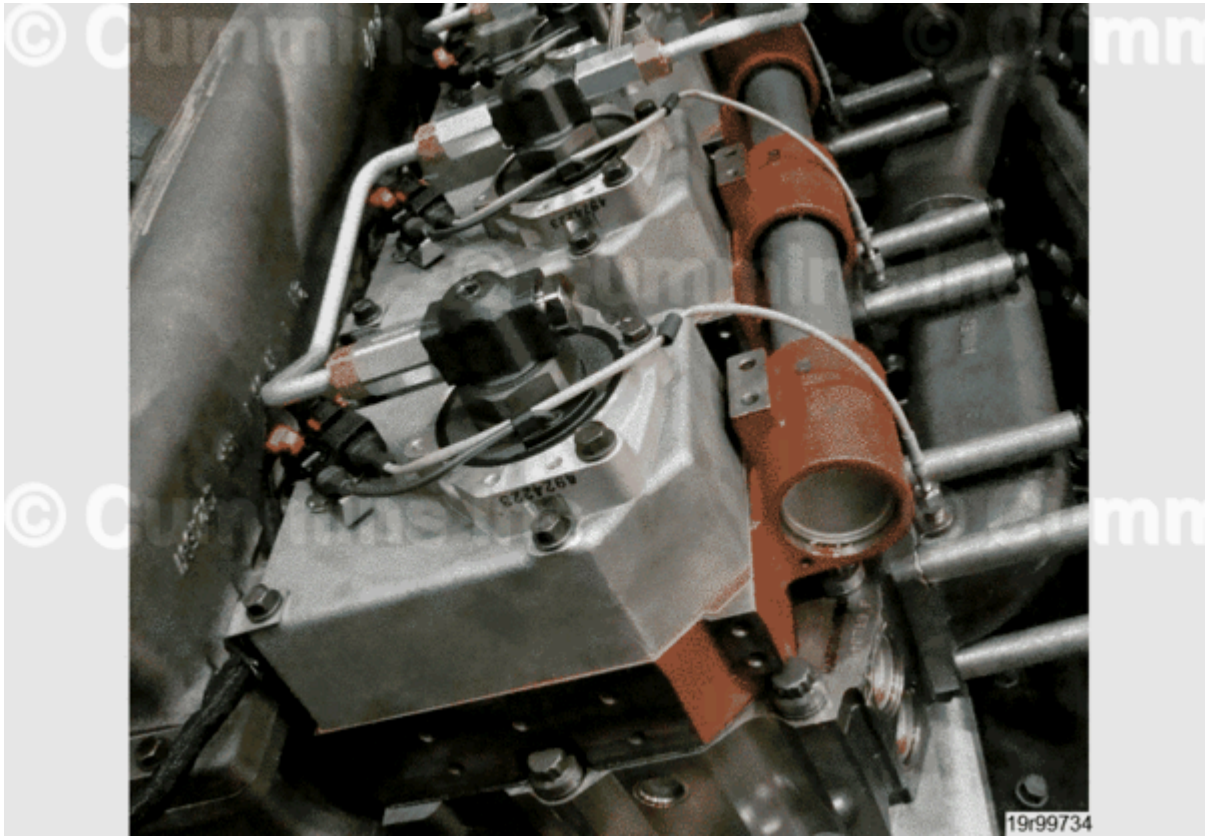


Figure 2, Existing EGTS Without Extended Body.

New Thermocouple Exhaust Gas Temperature Sensors

Four new thermocouple EGTS are required to properly route the sensors from the exhaust manifold and avoid interfering with other components such as water and exhaust manifolds, turbochargers, etc. at the top of engines. The inconnel of the thermocouple EGTS has a higher temperature rating is much longer than the thermistor EGTS. The thermocouple EGTS have dark orange cables.

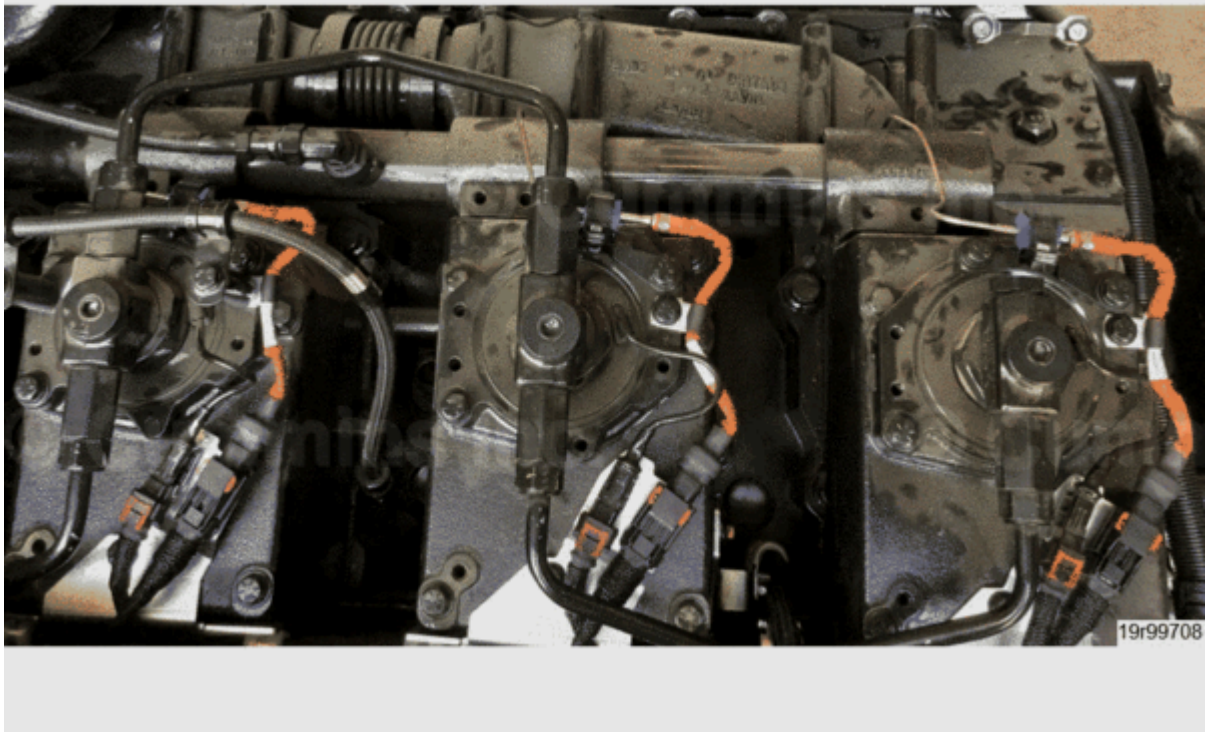


Figure 3, New EGTS Routing.

See Table 1 below to identify what EGTS part number is compatible with a cylinder location.

Table 1, EGTS Cylinder Locator

EGTS Part Number	Cylinder Number Locations
5462023	1
5462024	2, 3, 4, 5, 6, 7, 10, 11, 12, 13
5462025	8, 9
5462026	14, 15, 16

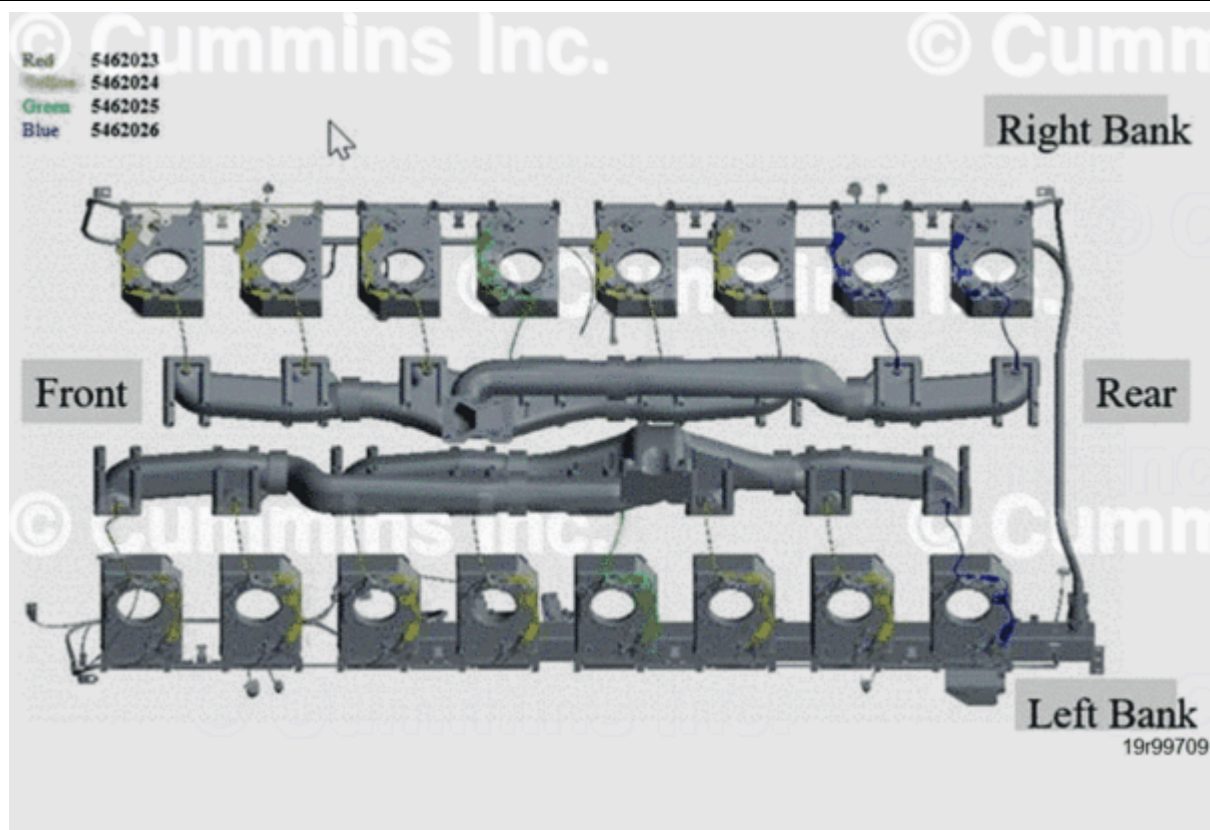


Figure 4, EGTS Part Number Locations by Cylinder.

Compatible Left Bank and Right Bank Main Engine Wiring Harnesses

The main engine wiring harnesses now have an additional converter box connector breakout. This can be found on both the Left Bank (LB) and Right Bank (RB) main engine wiring harnesses.

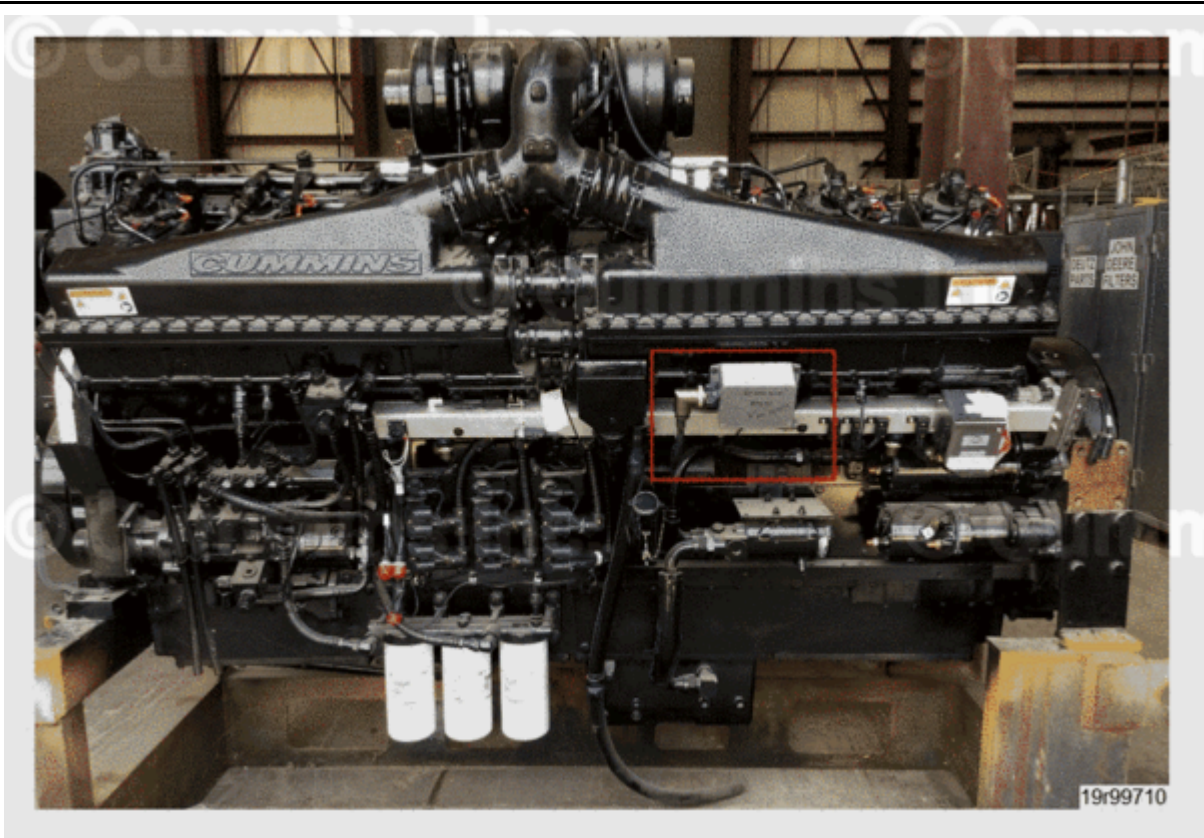


Figure 5, EGTS Converter Box.

Compatible Left Bank and Right Bank Injector/ EGTS Wiring Harnesses

The internal wires of the injector/ EGTS harnesses have been changed to wires which support the new thermocouples. The wiring harness connectors have been combined to one single bigger connector which connects to the main engine wiring harnesses.



Figure 6, New Wiring Harness Connections Points.



Figure 7, New Wiring Harness Connections Points.

Left Bank and Right Bank Converter Boxes and Mounting Brackets

Two converter boxes are required to house and protect the internal signal conversion modules. The internal signal conversion modules process the thermocouple EGTS's signals for the engine's ECMs. Both LB and RB converter boxes are mounted on the air intake manifold of the engine.

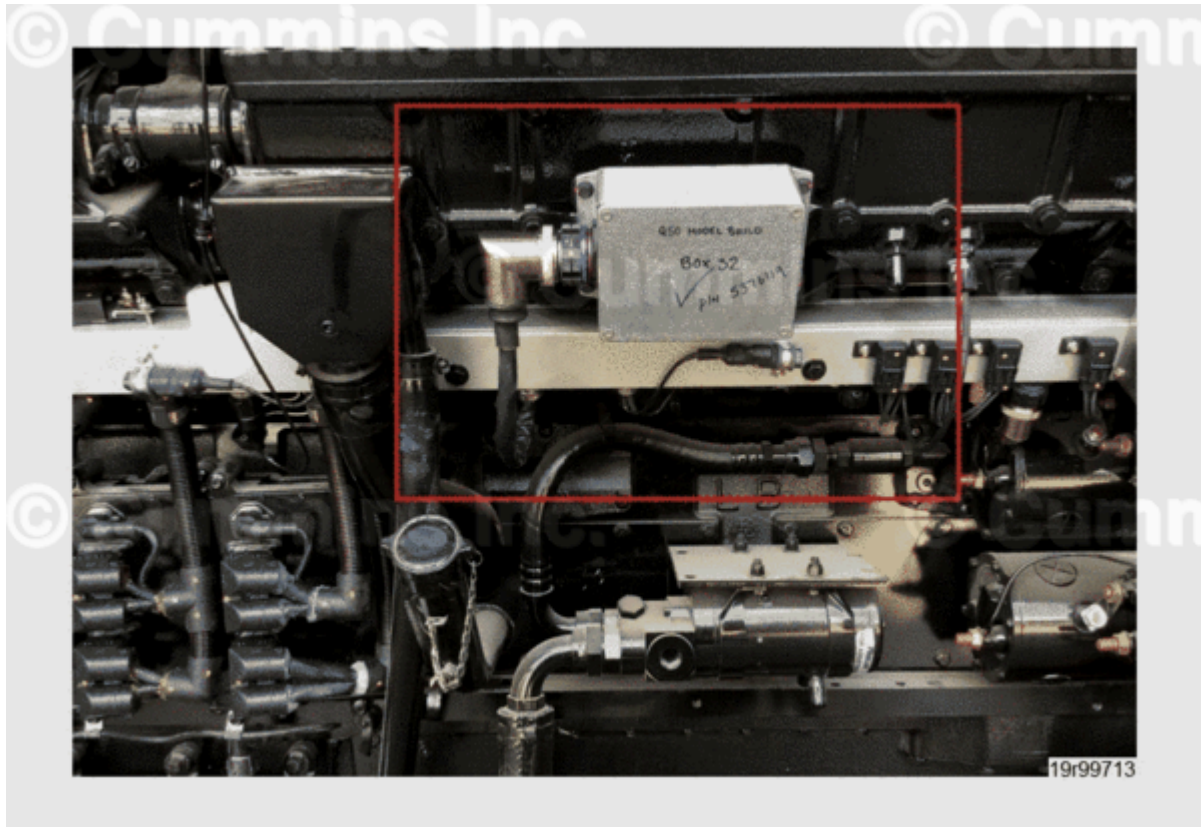


Figure 8, EGTS Converter Box.

Electronic Control Module (ECM) Calibrations

A revised ECM calibration is needed to recognize, transmit, and interpret the signals of the thermocouple sensors. A detail overview of the new ECM calibration will be covered in a separate TSB. See Technical Service Bulletin, Exhaust Gas Temperature (EGT) Sensor Types and Selection Strategy, TSB210137, for more details on the new ECM calibrations.

Service Instructions

For engines to use the thermocouple EGTS, the following hardware is required:

- Compatible left bank and right bank main engine harnesses.
- Compatible left bank and right bank injector/ EGTS harnesses.
- Left bank and right bank converter boxes and mounting brackets.
- Compatible electronic control module (ECM) calibrations.

Service Parts Availability

Service parts are available. See Table 2 for part numbers.

Table 2, Service Parts					
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Quantity Per Engine
Temperature, Sensor	4326918	No	No	5462023	1
				5462024	10
				5462025	2
				5462026	3
Captive Washer Cap Screw	-	-	-	3609817	16
Captive Washer Cap Screw	-	-	-	3015283	33
Wear Sleeve	-	-	-	5539796	16
Clip	-	-	-	4924291	16
Sensor Bracket	-	-	-	5376160	16
Bracket Support	-	-	-	5376161	16
Water Tube Bracket	-	-	-	5372025	4
LB Main Harness-Industrial	3649114	No	No	5538144	1
LB Main Harness – Industrial (LIEBHERR)	3649235	No	No	5538740	1
RB Main Harness - Industrial	3645547	No	No	5538145	1
LB Injector/EGTS Harness	5538957	No	No	5538148	2
RB Injector/EGTS Harness	5372878	No	No	5538146	2

Table 2, Service Parts					
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Quantity Per Engine
LB Main / Injector / EGTS Harness (AGGREKO)	3650849	No	No	6416926	1
RB Main / Injector / EGTS Harness (AGGREKO)	2880749	No	No	6416933	1
Fuel Filter Harness (Off Engine)	3643872	No	No	5538926	1
Fuel Filter Harness (On Engine)	3641274	No	No	5538937	1
Quadrant Harness Protection Bracket	-	-	-	5372029	4
Convertor Box Mounting Bracket	-	-	-	5538108	2
Sensor Bracket	-	-	-	5376160	16
Bracket Support	-	-	-	5376161	16
Analog Convertor Box	-	-	-	5572259	2

Part Compatibility

Thermocouple EGTS are **not** backwards compatible and are **not** able to be mixed with the thermistor EGTS on the same engine.

Production Status

Implemented for production. See Table 3.

Table 3, Production Information					
ESN First		Build Date ¹		Plant	
33222656		1 September 2020		Daventry Engine Plant	
¹ Engine build date can be found on engine dataplate.					
Publications Affected					
Table 4, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSK50 CM2150 MCRS	4022102	Exhaust Gas Temperature Sensor Converter	Refer to Procedure 019-450 (/qs3/pubsys2/xml/en/procedures/122/122-019-450.html)	19
Service Manual	QSK50 CM2150 MCRS	4022102	Exhaust Temperature Sensor	Refer to Procedure 019-013 (/qs3/pubsys2/xml/en/procedures/122/122-019-013.html)	19

Document History

Date	Details
2021-1-4	Module Created
2021-3-5	Updated Table 2.
2021-5-19	Updated Figure 2.
2021-7-1	Updated Description of Change section to reference TSB210137.
2023-5-11	Updated Table 2 part numbers.
2024-2-19	Updated Table 2 part numbers.

